FIELD THEORY

B.E. Electrical Engineering
Second Year – Second Semester
2022 Examination – Part II

Complete Semester-Style Solution
Compile-Safe LaTeX Code

Compile-Safe Note

This document is intentionally written without the `physics` package and without `tikz-3dplot`. So it avoids command conflicts such as:

```
Command \divergence already defined
```

and TikZ errors such as:

```
I do not know the key /tikz/tdplot_main_coords.
```

All diagrams are native 2D TikZ diagrams.

Contents

1 Question 1(a): Potential Gradient and Electric Field Intensity

What is potential gradient? How is it related to electric field intensity? Calculate electric field intensity at point $P(1, 0, 4)$ m due to $Q_1 = +0.24\,\mu$C at $(2, 4, 2)$ m and $Q_2 = -0.8\,\mu$C at $(4, 1, 0)$ m.

1.1 Potential Gradient

Electric potential V is a scalar quantity.

It varies from point to point in space.

The rate of change of potential with distance is called potential gradient.

In vector form, potential gradient is written as

$$\nabla V = \hat{\mathbf{x}}\frac{\partial V}{\partial x} + \hat{\mathbf{y}}\frac{\partial V}{\partial y} + \hat{\mathbf{z}}\frac{\partial V}{\partial z}.$$

Electric field intensity is the negative gradient of potential:

$$\boxed{\mathbf{E} = -\nabla V}$$

This negative sign means electric field points in the direction in which potential decreases most rapidly.

In one-dimensional form,

$$\boxed{E = -\frac{dV}{dl}}$$

1.2 Formula for Electric Field Due to Point Charge

Electric field at point P due to a point charge Q at position $\mathbf{r}_Q$ is

$$\boxed{\mathbf{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{R^3}\mathbf{R}}$$

where

$$\mathbf{R} = \mathbf{r}_P - \mathbf{r}_Q.$$

For air,

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9.$$

1.3 Field Due to Q_1

Given,

$$\mathbf{r}_P = (1, 0, 4)$$

and

$$\mathbf{r}_{Q_1} = (2, 4, 2).$$

Therefore,

$$\mathbf{R}_1 = (1-2)\hat{\mathbf{x}} + (0-4)\hat{\mathbf{y}} + (4-2)\hat{\mathbf{z}}.$$

$$\mathbf{R}_1 = -\hat{\mathbf{x}} - 4\hat{\mathbf{y}} + 2\hat{\mathbf{z}}.$$

Magnitude:

$$R_1 = \sqrt{(-1)^2 + (-4)^2 + (2)^2} = \sqrt{21}.$$

So,

$$R_1^3 = (\sqrt{21})^3 = 21\sqrt{21}.$$

Now,

$$\mathbf{E}_1 = 9 \times 10^9 \frac{0.24 \times 10^{-6}}{21\sqrt{21}}(-\hat{\mathbf{x}} - 4\hat{\mathbf{y}} + 2\hat{\mathbf{z}}).$$

$$9 \times 10^9 (0.24 \times 10^{-6}) = 2160.$$

Therefore,

$$\mathbf{E}_1 = \frac{2160}{21\sqrt{21}}(-\hat{\mathbf{x}} - 4\hat{\mathbf{y}} + 2\hat{\mathbf{z}}).$$

Numerically,

$$\mathbf{E}_1 = -22.45\hat{\mathbf{x}} - 89.78\hat{\mathbf{y}} + 44.89\hat{\mathbf{z}} \quad \text{V/m}.$$

1.4 Field Due to Q_2

Given,

$$\mathbf{r}_{Q_2} = (4, 1, 0).$$

Therefore,

$$\mathbf{R}_2 = (1-4)\hat{\mathbf{x}} + (0-1)\hat{\mathbf{y}} + (4-0)\hat{\mathbf{z}}.$$

$$\mathbf{R}_2 = -3\hat{\mathbf{x}} - \hat{\mathbf{y}} + 4\hat{\mathbf{z}}.$$

Magnitude:

$$R_2 = \sqrt{(-3)^2 + (-1)^2 + (4)^2} = \sqrt{26}.$$

Thus,

$$R_2^3 = 26\sqrt{26}.$$

Since

$$Q_2 = -0.8\,\mu C = -0.8 \times 10^{-6}\,C,$$

$$\mathbf{E}_2 = 9 \times 10^9 \frac{-0.8 \times 10^{-6}}{26\sqrt{26}}(-3\hat{\mathbf{x}} - \hat{\mathbf{y}} + 4\hat{\mathbf{z}}).$$

$$9 \times 10^9(-0.8 \times 10^{-6}) = -7200.$$

So,

$$\mathbf{E}_2 = \frac{-7200}{26\sqrt{26}}(-3\hat{\mathbf{x}} - \hat{\mathbf{y}} + 4\hat{\mathbf{z}}).$$

Numerically,

$$\mathbf{E}_2 = 162.93\hat{\mathbf{x}} + 54.31\hat{\mathbf{y}} - 217.24\hat{\mathbf{z}} \quad \text{V/m}.$$

1.5 Resultant Field at P

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2.$$

$$\mathbf{E} = (-22.45 + 162.93)\hat{\mathbf{x}} + (-89.78 + 54.31)\hat{\mathbf{y}} + (44.89 - 217.24)\hat{\mathbf{z}}.$$

$$\boxed{\mathbf{E} = 140.48\hat{\mathbf{x}} - 35.47\hat{\mathbf{y}} - 172.35\hat{\mathbf{z}} \quad \text{V/m}}$$

Magnitude:

$$|\mathbf{E}| = \sqrt{(140.48)^2 + (-35.47)^2 + (-172.35)^2}.$$

$$\boxed{|\mathbf{E}| = 225.16 \text{ V/m}}$$

$$\boxed{\mathbf{E} = 140.48\hat{\mathbf{x}} - 35.47\hat{\mathbf{y}} - 172.35\hat{\mathbf{z}} \quad \text{V/m}}$$

$$\boxed{|\mathbf{E}| = 225.16 \text{ V/m}}$$

2 Question 1(b): Ring Charge and Disc Charge

Consider a ring charge of radius $10\,\text{cm}$ and uniform charge density $+0.8\,\text{nC/m}$, and also a disc charge of radius $20\,\text{cm}$ and uniform charge density $\sigma\,\text{C/m}^2$. Both charges are placed in the xy-plane with centre at origin. If electric field intensity at a point of height $25\,\text{cm}$ on the z-axis is same due to the ring and disc charges individually, find the magnitude of σ. Relative permittivity of medium is 2.7.

2.1 Given Data

Ring radius:

$$a = 10\,\text{cm} = 0.10\,\text{m}.$$

Line charge density of ring:

$$\lambda = 0.8\,\text{nC/m} = 0.8 \times 10^{-9}\,\text{C/m}.$$

Disc radius:

$$R = 20\,\text{cm} = 0.20\,\text{m}.$$

Height of point on axis:

$$z = 25\,\text{cm} = 0.25\,\text{m}.$$

Relative permittivity:

$$\epsilon_r = 2.7.$$

So,

$$\epsilon = \epsilon_0 \epsilon_r.$$

2.2 Electric Field Due to Ring on Axis

Total charge on ring is

$$Q = \lambda(2\pi a).$$

Electric field on the axis of a uniformly charged ring is

$$E_{\text{ring}} = \frac{1}{4\pi\epsilon} \frac{Qz}{(z^2 + a^2)^{3/2}}.$$

Substitute $Q = 2\pi a \lambda$:

$$E_{\text{ring}} = \frac{1}{4\pi\epsilon} \frac{(2\pi a \lambda)z}{(z^2 + a^2)^{3/2}}.$$

$$\boxed{E_{\text{ring}} = \frac{a\lambda z}{2\epsilon(z^2+a^2)^{3/2}}}$$

2.3 Electric Field Due to Uniform Disc on Axis

Electric field on the axis of a uniformly charged disc is

$$\boxed{E_{\text{disc}} = \frac{\sigma}{2\epsilon}\left[1 - \frac{z}{\sqrt{z^2+R^2}}\right]}$$

2.4 Condition Given in Problem

The electric fields are equal individually at the same point:

$$E_{\text{ring}} = E_{\text{disc}}.$$

So,

$$\frac{a\lambda z}{2\epsilon(z^2+a^2)^{3/2}} = \frac{\sigma}{2\epsilon}\left[1 - \frac{z}{\sqrt{z^2+R^2}}\right].$$

Cancel $\frac{1}{2\epsilon}$ from both sides:

$$\frac{a\lambda z}{(z^2+a^2)^{3/2}} = \sigma\left[1 - \frac{z}{\sqrt{z^2+R^2}}\right].$$

Therefore,

$$\boxed{\sigma = \frac{a\lambda z}{(z^2+a^2)^{3/2}\left[1 - \frac{z}{\sqrt{z^2+R^2}}\right]}}$$

2.5 Numerical Calculation

Substitute:

$$a = 0.10, \qquad \lambda = 0.8\times10^{-9}, \qquad z = 0.25, \qquad R = 0.20.$$

First,

$$z^2 + a^2 = (0.25)^2 + (0.10)^2 = 0.0625 + 0.0100 = 0.0725.$$

$$(z^2+a^2)^{3/2} = (0.0725)^{3/2}.$$

Also,

$$\sqrt{z^2+R^2} = \sqrt{(0.25)^2+(0.20)^2} = \sqrt{0.1025} = 0.32016.$$

$$1 - \frac{z}{\sqrt{z^2 + R^2}} = 1 - \frac{0.25}{0.32016} = 1 - 0.78087 = 0.21913.$$

Using the formula,

$$\sigma \approx 4.68 \times 10^{-9}\,\mathrm{C/m^2}.$$

$$\boxed{\sigma \approx 4.68\,\mathrm{nC/m^2}}$$

2.6 Important Observation

The permittivity cancels because both fields are in the same medium.
So the final value of σ is independent of ϵ_r for this equality condition.

$$\boxed{\sigma = 4.68 \times 10^{-9}\,\mathrm{C/m^2} = 4.68\,\mathrm{nC/m^2}}$$

3 Question 2(a): Integral Form of Gauss Law

State and prove the integral form of Gauss law with proper explanation.

3.1 Statement

Gauss law states that total electric flux through any closed surface is equal to the total charge enclosed by that surface.
In terms of electric flux density,

$$\boxed{\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}}}$$

This is the integral form of Gauss law.
If the charge is distributed with volume charge density ρ_v, then

$$Q_{\text{enc}} = \iiint_V \rho_v \, dv.$$

Therefore,

$$\boxed{\oint_S \mathbf{D} \cdot d\mathbf{S} = \iiint_V \rho_v \, dv}$$

3.2 Proof from Point Charge

Consider a point charge Q at the centre of a spherical surface of radius r.
For a point charge,

$$\mathbf{E} = \frac{Q}{4\pi\epsilon r^2}\hat{\mathbf{r}}.$$

Since

$$\mathbf{D} = \epsilon \mathbf{E},$$

$$\mathbf{D} = \frac{Q}{4\pi r^2}\hat{\mathbf{r}}.$$

For a spherical surface,

$$d\mathbf{S} = r^2 \sin\theta \, d\theta \, d\phi \, \hat{\mathbf{r}}.$$

So,

$$\mathbf{D} \cdot d\mathbf{S} = \frac{Q}{4\pi r^2} r^2 \sin\theta \, d\theta \, d\phi.$$

$$= \frac{Q}{4\pi} \sin\theta \, d\theta \, d\phi.$$

Therefore,

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_0^{2\pi} \int_0^{\pi} \frac{Q}{4\pi} \sin\theta \, d\theta \, d\phi.$$

$$= \frac{Q}{4\pi}(2\pi)(2).$$

$$= Q.$$

Hence,

$$\boxed{\oint_S \mathbf{D} \cdot d\mathbf{S} = Q}$$

For any closed surface enclosing charge Q, the same result is obtained because total flux depends only on enclosed charge, not on the shape of the surface.

3.3 Differential Form Connection

Using divergence theorem,

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \iiint_V \nabla \cdot \mathbf{D} \, dv.$$

But Gauss law says,

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \iiint_V \rho_v \, dv.$$

Therefore,

$$\iiint_V \nabla \cdot \mathbf{D} \, dv = \iiint_V \rho_v \, dv.$$

Since the volume is arbitrary,

$$\boxed{\nabla \cdot \mathbf{D} = \rho_v}$$

Integral form:

$$\boxed{\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}}}$$

Differential form:

$$\boxed{\nabla \cdot \mathbf{D} = \rho_v}$$

4 Question 2(b): Electric Field on Conductor Surface and Mechanical Pressure

Show that electric field intensity on the conductor surface is half of the electric field intensity just off the conductor surface. Also, deduce an expression for the mechanical pressure on conductor surface.

4.1 Field Just Outside and Inside a Conductor

For a conductor in electrostatic equilibrium,

$$\mathbf{E}_{\text{inside}} = 0.$$

Let surface charge density be

$$\rho_s.$$

At the conductor surface, boundary condition gives

$$D_n = \rho_s.$$

Since

$$D_n = \epsilon E_{\text{out}},$$

we get

$$\boxed{E_{\text{out}} = \frac{\rho_s}{\epsilon}}$$

This is the electric field intensity just outside the conductor surface.

4.2 Electric Field Acting on Surface Charge

The electric field exactly acting on the surface charge is the average of the field just outside and just inside the conductor.

$$E_{\text{surface}} = \frac{E_{\text{out}} + E_{\text{inside}}}{2}.$$

Since

$$E_{\text{inside}} = 0,$$

$$E_{\text{surface}} = \frac{E_{\text{out}}}{2}.$$

Thus,

$$\boxed{E_{\text{surface}} = \frac{1}{2}E_{\text{out}}}$$

Using

$$E_{\text{out}} = \frac{\rho_s}{\epsilon},$$

$$\boxed{E_{\text{surface}} = \frac{\rho_s}{2\epsilon}}$$

4.3 Mechanical Pressure

Force on charge is

$$F = qE.$$

Force per unit area is pressure.
Surface charge per unit area is ρ_s.
Therefore,

$$p = \rho_s E_{\text{surface}}.$$

Substitute

$$E_{\text{surface}} = \frac{\rho_s}{2\epsilon}.$$

$$p = \rho_s \frac{\rho_s}{2\epsilon}.$$

$$\boxed{p = \frac{\rho_s^2}{2\epsilon}}$$

Since

$$\rho_s = \epsilon E_{\text{out}},$$

$$p = \frac{(\epsilon E_{\text{out}})^2}{2\epsilon}.$$

Therefore,

$$\boxed{p = \frac{1}{2}\epsilon E_{\text{out}}^2}$$

This pressure acts normally outward on the conductor surface.

$$\boxed{E_{\text{surface}} = \frac{1}{2}E_{\text{out}}}$$

$$\boxed{p = \frac{\rho_s^2}{2\epsilon} = \frac{1}{2}\epsilon E_{\text{out}}^2}$$

5 Question 3(a): Field Utilization Factor of Cylindrical Coaxial Insulation

What do you mean by field utilization factor? Calculate the field utilization factor of a cylindrical coaxial insulation.

5.1 Definition

Field utilization factor is the ratio of average electric field intensity to maximum electric field intensity.

It is denoted by η.

$$\boxed{\eta = \frac{E_{\text{avg}}}{E_{\text{max}}}}$$

It indicates how uniformly the insulation is utilized.

If η is close to 1, electric field distribution is more uniform.

If η is small, field is highly non-uniform.

5.2 Cylindrical Coaxial Insulation

Consider a coaxial insulation between two cylindrical conductors.

Let inner conductor radius be a.

Let outer conductor inner radius be b.

Let potential difference be V.

Electric field at radius r is

$$E(r) = \frac{\lambda}{2\pi\epsilon r}.$$

The potential difference is

$$V = \int_a^b E(r)\, dr.$$

$$V = \int_a^b \frac{\lambda}{2\pi\epsilon r}\, dr.$$

$$V = \frac{\lambda}{2\pi\epsilon} \ln \frac{b}{a}.$$

Therefore,

$$\frac{\lambda}{2\pi\epsilon} = \frac{V}{\ln(b/a)}.$$

So,

$$E(r) = \frac{V}{r \ln(b/a)}.$$

Maximum electric field occurs at inner conductor surface $r = a$:

$$E_{\text{max}} = \frac{V}{a \ln(b/a)}.$$

Average electric field across insulation thickness is

$$E_{\text{avg}} = \frac{V}{b - a}.$$

Therefore, field utilization factor is

$$\eta = \frac{E_{\text{avg}}}{E_{\text{max}}}.$$

$$\eta = \frac{\frac{V}{b-a}}{\frac{V}{a \ln(b/a)}}.$$

$$\boxed{\eta = \frac{a \ln(b/a)}{b - a}}$$

$$\boxed{\eta = \frac{E_{\text{avg}}}{E_{\text{max}}} = \frac{a \ln(b/a)}{b - a}}$$

6 Question 3(b): Two-Layer Single-Core Cable

A single-core lead sheathed cable has a core of 20 cm diameter and two layers of different dielectric media of thickness 8 cm each. The relative permittivities are 4.5 for inner dielectric and 2.4 for outer dielectric. Calculate the maximum and minimum electric field intensities and their locations within the cable when the potential difference between the core and sheath is 33 kV.

6.1 Given Data

Core diameter:

$$20\,\mathrm{cm}.$$

Therefore core radius:

$$a = 10\,\mathrm{cm} = 0.10\,\mathrm{m}.$$

Thickness of each dielectric layer:

$$8\,\mathrm{cm} = 0.08\,\mathrm{m}.$$

Radius at interface:

$$b = 0.10 + 0.08 = 0.18\,\mathrm{m}.$$

Outer sheath radius:

$$c = 0.18 + 0.08 = 0.26\,\mathrm{m}.$$

Relative permittivity of inner layer:

$$\epsilon_{r1} = 4.5.$$

Relative permittivity of outer layer:

$$\epsilon_{r2} = 2.4.$$

Voltage:

$$V = 33\,\mathrm{kV} = 33000\,\mathrm{V}.$$

6.2 Electric Field in Two Layers

Let

$$K = \frac{\lambda}{2\pi\epsilon_0}.$$

Then in inner dielectric,

$$E_1(r) = \frac{K}{\epsilon_{r1} r}.$$

In outer dielectric,

$$E_2(r) = \frac{K}{\epsilon_{r2} r}.$$

Potential difference is

$$V = \int_a^b E_1(r)\, dr + \int_b^c E_2(r)\, dr.$$

$$V = K\left[\frac{1}{\epsilon_{r1}} \ln\frac{b}{a} + \frac{1}{\epsilon_{r2}} \ln\frac{c}{b}\right].$$

Therefore,

$$K = \frac{V}{\frac{1}{\epsilon_{r1}} \ln\frac{b}{a} + \frac{1}{\epsilon_{r2}} \ln\frac{c}{b}}.$$

6.3 Calculate K

$$K = \frac{33000}{\frac{1}{4.5} \ln\frac{0.18}{0.10} + \frac{1}{2.4} \ln\frac{0.26}{0.18}}.$$

$$\ln\frac{0.18}{0.10} = \ln(1.8) = 0.5878.$$

$$\ln\frac{0.26}{0.18} = \ln(1.4444) = 0.3677.$$

Thus denominator is

$$\frac{0.5878}{4.5} + \frac{0.3677}{2.4} = 0.1306 + 0.1532 = 0.2838.$$

So,

$$K = \frac{33000}{0.2838} = 116263.5.$$

6.4 Maximum and Minimum Fields in Each Layer

In the inner layer,

$$E_1(r) = \frac{K}{4.5r}.$$

Maximum occurs at $r = a = 0.10\,\mathrm{m}$:

$$E_{1\,\max} = \frac{116263.5}{4.5(0.10)} = 258363.4\,\mathrm{V/m}.$$

$$\boxed{E_{1\,\max} = 258.36\,\mathrm{kV/m}}$$

Minimum in inner layer occurs at $r = b = 0.18\,\mathrm{m}$ just inside the interface:

$$E_{1\,\min} = \frac{116263.5}{4.5(0.18)} = 143535.2\,\mathrm{V/m}.$$

$$\boxed{E_{1\,\min} = 143.54\,\mathrm{kV/m}}$$

In the outer layer,

$$E_2(r) = \frac{K}{2.4r}.$$

Maximum in outer layer occurs at $r = b = 0.18\,\mathrm{m}$ just outside the interface:

$$E_{2\,\max} = \frac{116263.5}{2.4(0.18)} = 269128.6\,\mathrm{V/m}.$$

$$\boxed{E_{2\,\max} = 269.13\,\mathrm{kV/m}}$$

Minimum in outer layer occurs at $r = c = 0.26\,\mathrm{m}$:

$$E_{2\,\min} = \frac{116263.5}{2.4(0.26)} = 186319.8\,\mathrm{V/m}.$$

$$\boxed{E_{2\,\min} = 186.32\,\mathrm{kV/m}}$$

6.5 Overall Maximum and Minimum

Compare all values:

$$E_{1\,\max} = 258.36\,\mathrm{kV/m}$$

$$E_{1\,\min} = 143.54\,\mathrm{kV/m}$$

$$E_{2\,\max} = 269.13\,\mathrm{kV/m}$$

$$E_{2\,\min} = 186.32\,\mathrm{kV/m}$$

Overall maximum is

$$\boxed{E_{\max} = 269.13\,\mathrm{kV/m}}$$

at

$$\boxed{r = 0.18\,\mathrm{m}}$$

just outside the interface in the outer dielectric.

Overall minimum is

$$\boxed{E_{\text{min}} = 143.54\,\text{kV/m}}$$

at

$$\boxed{r = 0.18\,\text{m}}$$

just inside the interface in the inner dielectric.

$$\boxed{E_{\text{max}} = 269.13\,\text{kV/m} \quad \text{at } r = 18\,\text{cm, just outside interface}}$$

$$\boxed{E_{\text{min}} = 143.54\,\text{kV/m} \quad \text{at } r = 18\,\text{cm, just inside interface}}$$

7 Question 4(a): Finite Difference Method for Given Grid

For the two-dimensional system with equal nodal distances shown in Fig. 1, write the FDM equations for the unknown node potentials. Derive the formula used.

7.1 Formula Used

For a two-dimensional potential problem with equal nodal spacing,

$$\nabla^2 V = 0.$$

Using central difference approximation,

$$\frac{V_E - 2V_0 + V_W}{h^2} + \frac{V_N - 2V_0 + V_S}{h^2} = 0.$$

Since nodal distances are equal,

$$V_E + V_W + V_N + V_S - 4V_0 = 0.$$

Therefore,

$$\boxed{V_0 = \frac{V_E + V_W + V_N + V_S}{4}}$$

This is the five-point finite difference formula.

7.2 Grid Interpretation

Unknown node potentials are

$$V_1, \quad V_2, \quad V_3, \quad V_4.$$

Boundary values from the figure are:
Top boundary:

$$100\,\text{kV}.$$

Left side upper boundary:

$$78\,\text{kV}.$$

Left side lower boundary:

$$41\,\text{kV}.$$

Right side upper boundary:

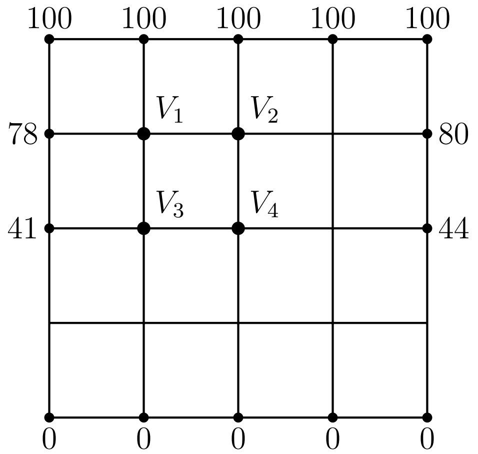
$$80\,\text{kV}.$$

Right side lower boundary:

$$44\,\text{kV}.$$

Bottom boundary:

$$0\,\text{kV}.$$



FDM grid with four unknown nodes

Figure 1: FDM grid recreated from the question.

7.3 FDM Equation for Node 1

For node 1:

$$V_1 = \frac{78 + 100 + V_2 + V_3}{4}.$$

Thus,

$$4V_1 - V_2 - V_3 = 178.$$

$$\boxed{4V_1 - V_2 - V_3 = 178}$$

7.4 FDM Equation for Node 2

For node 2:

$$V_2 = \frac{V_1 + 100 + 80 + V_4}{4}.$$

Thus,

$$\boxed{-V_1 + 4V_2 - V_4 = 180}$$

7.5 FDM Equation for Node 3

For node 3:

$$V_3 = \frac{41 + V_1 + V_4 + 0}{4}.$$

Thus,

$$\boxed{-V_1 + 4V_3 - V_4 = 41}$$

7.6 FDM Equation for Node 4

For node 4:

$$V_4 = \frac{V_3 + V_2 + 44 + 0}{4}.$$

Thus,

$$\boxed{-V_2 - V_3 + 4V_4 = 44}$$

7.7 Solution of Simultaneous Equations

The equations are:

$$4V_1 - V_2 - V_3 = 178.$$

$$-V_1 + 4V_2 - V_4 = 180.$$

$$-V_1 + 4V_3 - V_4 = 41.$$

$$-V_2 - V_3 + 4V_4 = 44.$$

Solving,

$$\boxed{V_1 = 72.17\,\mathrm{kV}}$$

$$\boxed{V_2 = 72.71\,\mathrm{kV}}$$

$$\boxed{V_3 = 37.96\,\mathrm{kV}}$$

$$\boxed{V_4 = 38.67\,\mathrm{kV}}$$

$$\boxed{V_1 = 72.17\,\mathrm{kV}, \quad V_2 = 72.71\,\mathrm{kV}, \quad V_3 = 37.96\,\mathrm{kV}, \quad V_4 = 38.67\,\mathrm{kV}}$$

8 Question 4(b): Point Charge Above Conducting Plane

A point charge of $1\,\mathrm{mC}$ is placed at $(0, 0, 1)$ m. An infinitely long conducting plane is there at $z = 0$. Find the electric potential and electric stresses at point $(3, 4, 1)$ m. The medium is air.

8.1 Method of Images

A grounded infinite conducting plane at $z = 0$ can be replaced by an image charge.
Real charge:

$$Q = 1\,\mathrm{mC} = 10^{-3}\,\mathrm{C}$$

at

$$(0, 0, 1).$$

Image charge:

$$-Q = -10^{-3}\,\mathrm{C}$$

at

$$(0, 0, -1).$$

Observation point:

$$P(3, 4, 1).$$

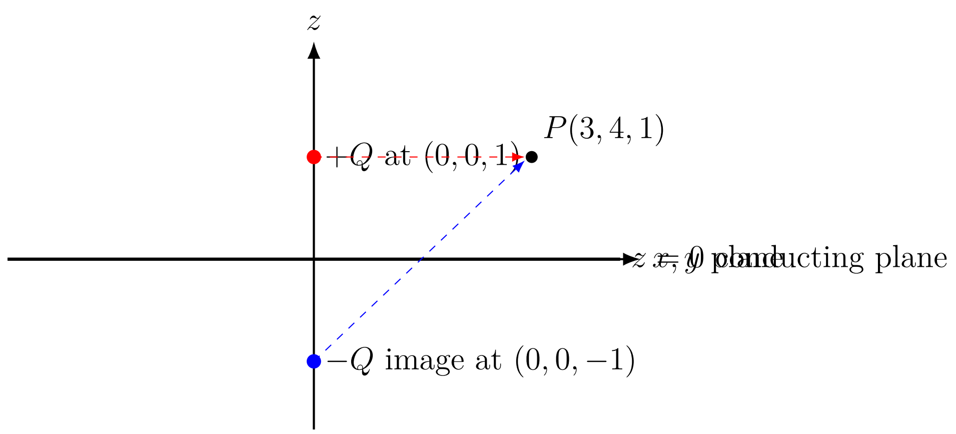


Figure 2: Real charge and image charge for grounded conducting plane.

8.2 Distances

Distance from real charge to P:

$$R_1 = \sqrt{(3-0)^2 + (4-0)^2 + (1-1)^2}.$$

$$R_1 = \sqrt{9 + 16 + 0} = 5\,\text{m}.$$

Distance from image charge to P:

$$R_2 = \sqrt{(3-0)^2 + (4-0)^2 + (1+1)^2}.$$

$$R_2 = \sqrt{9 + 16 + 4} = \sqrt{29} = 5.385\,\text{m}.$$

8.3 Electric Potential

Potential at P is

$$V = \frac{1}{4\pi\epsilon_0}\left[\frac{Q}{R_1} - \frac{Q}{R_2}\right].$$

Using

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9,$$

$$V = 9 \times 10^9 (10^{-3})\left[\frac{1}{5} - \frac{1}{\sqrt{29}}\right].$$

$$V = 9 \times 10^6 (0.2000 - 0.1857).$$

$$V = 9 \times 10^6 (0.01430).$$

$$\boxed{V = 1.287 \times 10^5\,\text{V}}$$

$$\boxed{V = 128.74\,\text{kV}}$$

8.4 Electric Field Components

Vector from real charge to P:

$$\mathbf{R}_1 = 3\hat{\mathbf{x}} + 4\hat{\mathbf{y}} + 0\hat{\mathbf{z}}.$$

Vector from image charge to P:

$$\mathbf{R}_2 = 3\hat{\mathbf{x}} + 4\hat{\mathbf{y}} + 2\hat{\mathbf{z}}.$$

Electric field is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} Q\left[\frac{\mathbf{R}_1}{R_1^3} - \frac{\mathbf{R}_2}{R_2^3}\right].$$

Substituting,

$$\mathbf{E} = 9 \times 10^9 (10^{-3}) \left[\frac{3\hat{\mathbf{x}} + 4\hat{\mathbf{y}}}{5^3} - \frac{3\hat{\mathbf{x}} + 4\hat{\mathbf{y}} + 2\hat{\mathbf{z}}}{(\sqrt{29})^3} \right].$$

Numerically,

$$\boxed{\mathbf{E} = 43111.24\hat{\mathbf{x}} + 57481.65\hat{\mathbf{y}} - 115259.18\hat{\mathbf{z}} \quad \mathrm{V/m}}$$

Magnitude:

$$|\mathbf{E}| = 135821.19\,\mathrm{V/m}.$$

$$\boxed{|\mathbf{E}| = 135.82\,\mathrm{kV/m}}$$

8.5 Meaning of Electric Stress

Here electric stresses mean the electric field intensity components at the point.
So,

$$E_x = 43.11\,\mathrm{kV/m},$$

$$E_y = 57.48\,\mathrm{kV/m},$$

$$E_z = -115.26\,\mathrm{kV/m}.$$

$$\boxed{V = 128.74\,\mathrm{kV}}$$

$$\boxed{\mathbf{E} = 43.11\hat{\mathbf{x}} + 57.48\hat{\mathbf{y}} - 115.26\hat{\mathbf{z}} \quad \mathrm{kV/m}}$$

9 Question 5(a): Uniqueness Theorem for Insulating Material

State Uniqueness Theorem for an insulating material. Prove the theorem for calculation of electric potential.

9.1 Statement

For a given insulating region, if the charge distribution and boundary conditions are specified, then the electric potential solution is unique.

For a charge-free insulating region, if boundary potential is specified, solution of Laplace equation is unique.

That means two different correct potential functions cannot satisfy the same field equation and the same boundary conditions.

9.2 Proof

Let two solutions exist:

$$V_1$$

and

$$V_2.$$

Both satisfy the same equation in the insulating region.

For a charge-free region,

$$\nabla^2 V_1 = 0$$

and

$$\nabla^2 V_2 = 0.$$

Both satisfy the same boundary condition on surface S.

Define

$$V_d = V_1 - V_2.$$

Then,

$$\nabla^2 V_d = \nabla^2 V_1 - \nabla^2 V_2.$$

$$\nabla^2 V_d = 0.$$

Also, on the boundary, since $V_1 = V_2$,

$$V_d = 0.$$

Use Green's first identity:

$$\iiint_V \left[|\nabla V_d|^2 + V_d \nabla^2 V_d \right] dv = \oint_S V_d \frac{\partial V_d}{\partial n} dS.$$

Since

$$\nabla^2 V_d = 0$$

and on boundary

$$V_d = 0,$$

the equation becomes

$$\iiint_V |\nabla V_d|^2 dv = 0.$$

The integrand

$$|\nabla V_d|^2$$

cannot be negative.
Therefore,

$$|\nabla V_d|^2 = 0.$$

So,

$$\nabla V_d = 0.$$

Therefore V_d is constant throughout the region.
Since

$$V_d = 0$$

on boundary, the constant must be zero.
Hence,

$$V_d = 0$$

everywhere.
Therefore,

$$V_1 = V_2.$$

So the solution is unique.

9.3 General Insulating Material Form

For non-uniform insulating material, governing equation is

$$\nabla \cdot (\epsilon \nabla V) = 0$$

in charge-free region.
Following similar energy integral method gives

$$\iiint_V \epsilon |\nabla V_d|^2 dv = 0.$$

Since

$$\epsilon > 0,$$

we get

$$\nabla V_d = 0.$$

Therefore solution is again unique.

$\boxed{\text{For specified boundary conditions, electric potential in an insulating region is unique.}}$

10 Question 5(b): Electric Dipole and Field Components

What is electric dipole? Calculate the electric field intensity components at any point due to an electric dipole.

10.1 Definition of Electric Dipole

An electric dipole consists of two equal and opposite point charges separated by a small distance. Let charges be

$$+Q$$

and

$$-Q.$$

Let separation distance be

$$2d.$$

Dipole moment is defined as

$$\boxed{\mathbf{p} = Q(2d)\hat{\mathbf{p}}}$$

where $\hat{\mathbf{p}}$ is directed from negative charge to positive charge.
Magnitude of dipole moment is

$$\boxed{p = 2Qd}$$

10.2 Potential Due to Dipole

For a point $P(r, \theta)$ far from the dipole,

$$r \gg d.$$

The electric potential due to an ideal dipole is

$$\boxed{V = \frac{p\cos\theta}{4\pi\epsilon r^2}}$$

Here θ is the angle between position vector $\mathbf{r}$ and dipole axis.

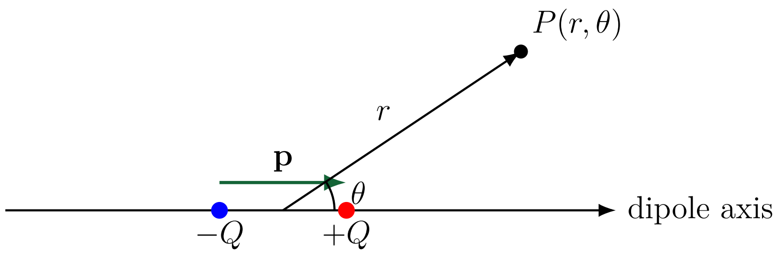


Figure 3: Electric dipole and observation point.

10.3 Electric Field Components

In spherical coordinates,

$$\mathbf{E} = E_r\hat{\mathbf{r}} + E_\theta\hat{\theta}.$$

The components are obtained from potential:

$$E_r = -\frac{\partial V}{\partial r}$$

and

$$E_\theta = -\frac{1}{r}\frac{\partial V}{\partial \theta}.$$

Given,

$$V = \frac{p\cos\theta}{4\pi\epsilon r^2}.$$

10.4 Radial Component

$$E_r = -\frac{\partial}{\partial r}\left[\frac{p\cos\theta}{4\pi\epsilon r^2}\right].$$

Since

$$\frac{\partial}{\partial r}(r^{-2}) = -2r^{-3},$$

$$E_r = -\left[\frac{p\cos\theta}{4\pi\epsilon}(-2r^{-3})\right].$$

Thus,

$$\boxed{E_r = \frac{2p\cos\theta}{4\pi\epsilon r^3}}$$

10.5 Transverse Component

$$E_\theta = -\frac{1}{r}\frac{\partial}{\partial\theta}\left[\frac{p\cos\theta}{4\pi\epsilon r^2}\right].$$

$$\frac{\partial}{\partial\theta}(\cos\theta) = -\sin\theta.$$

So,

$$E_\theta = -\frac{1}{r}\left[\frac{p}{4\pi\epsilon r^2}(-\sin\theta)\right].$$

Therefore,

$$\boxed{E_\theta = \frac{p\sin\theta}{4\pi\epsilon r^3}}$$

10.6 Total Field

Therefore,

$$\boxed{\mathbf{E} = \frac{p}{4\pi\epsilon r^3}\left(2\cos\theta\,\hat{\mathbf{r}} + \sin\theta\,\hat{\theta}\right)}$$

10.7 Special Cases

On axial line,

$$\theta = 0.$$

So,

$$E_r = \frac{2p}{4\pi\epsilon r^3}$$

and

$$E_\theta = 0.$$

On equatorial line,

$$\theta = 90^\circ.$$

So,

$$E_r = 0$$

and

$$E_\theta = \frac{p}{4\pi\epsilon r^3}.$$

$$\boxed{E_r = \frac{2p\cos\theta}{4\pi\epsilon r^3}}$$

$$\boxed{E_\theta = \frac{p\sin\theta}{4\pi\epsilon r^3}}$$

$$\boxed{\mathbf{E} = \frac{p}{4\pi\epsilon r^3}\left(2\cos\theta\,\hat{\mathbf{r}} + \sin\theta\,\hat{\theta}\right)}$$

11 Final Formula Sheet

$$\boxed{\mathbf{E} = -\nabla V}$$

$$\boxed{\mathbf{E} = \frac{1}{4\pi\epsilon}\frac{Q}{R^3}\mathbf{R}}$$

$$\boxed{E_{\text{ring}} = \frac{a\lambda z}{2\epsilon(z^2 + a^2)^{3/2}}}$$

$$\boxed{E_{\text{disc}} = \frac{\sigma}{2\epsilon}\left[1 - \frac{z}{\sqrt{z^2 + R^2}}\right]}$$

$$\boxed{\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}}}$$

$$\boxed{p = \frac{\rho_s^2}{2\epsilon} = \frac{1}{2}\epsilon E^2}$$

$$\boxed{\eta = \frac{a\ln(b/a)}{b - a}}$$

$$\boxed{V_0 = \frac{V_E + V_W + V_N + V_S}{4}}$$

$$\boxed{\textit{Cylindrical cable}: \quad E(r) = \frac{V}{r\ln(b/a)}}$$

$$\boxed{\textit{Dipole}: \quad V = \frac{p\cos\theta}{4\pi\epsilon r^2}}$$

$$\boxed{\textit{Dipole}: \quad E_r = \frac{2p\cos\theta}{4\pi\epsilon r^3}, \qquad E_\theta = \frac{p\sin\theta}{4\pi\epsilon r^3}}$$